

UNIVERSITY OF CALGARY

HETEROPLASMY IN HUMAN MITOCHONDRIAL GENOME MUTATION RATES
DIFFERENCES.

by

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A THESIS

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Abstract

From current literature, it is known that the mutation rate of the mitochondrial genome at a pedigree level is different from the evolution rate calculated from phylogeny-based analyses. While a definitive solution to this problem is not known, many explanations have been proposed. Such hypotheses include negative selection on the mitochondrial genome; high mutational regions of the genome known as “hotspots”; or the inflation of pedigree-based mutation rate estimates due to difficulties in discerning a mutation allele from an inherited allele in heteroplasmic samples. The main objective of this project is to evaluate the potential hypotheses for the differential rates of evolution between the two analysis methods and whether previously proposed models can accurately depict mitochondrial evolution. We curated a new dataset to calculate a pedigree-based mutation rate estimate with no heteroplasmy restriction. We observed a 14.9-to-22.4-fold higher calculated rate estimate compared to the literature rate estimates. Using a heteroplasmy filter, we observed that the rate estimate approach the high-end literature rate estimate of 2.6 mutations/site/million years. We also observed a mutation gradient within our curated dataset with “hotspot” regions for mutations aligning with hyper variable regions observed in previous literature. We concluded that pedigree mutation rate estimates are inflated in heteroplasmic samples and previously proposed models of the mitochondrial genome evolution need to account for heteroplasmy. We observed variation in our dataset according to the tissue sample which suggests that pedigree-based estimates are affected by the somatic mutation processes and should be considered in future studies.

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Reflection

Throughout this year working on this project, the major thing I learned was the value of time management and working diligently on a well-thought-out plan or schedule. Allowing myself to allocate certain hours in the day to thesis work, but more importantly, time for myself to de-stress and avoid burnout, is something I learned the importance of this year. Mainly finding motivation to work was a challenge during certain points in this semester. Obviously, life gets in the way even with a well-thought-out plan sometimes, and adjustments had to be made. With that being said, I learned about how I respond to hardships and how it makes me feel, which led to learning ways that help me bounce back from a slump. Pertaining to research, I have been able to experience start to finish the process of a research project that essentially was entirely my doing. This experience was intimidating and pushed me the most I have ever been pushed. Throughout this project, I was able to recognize skills that we learned in previous classes, such as the presentations, data collection and paper writing. Being able to apply these skills that I have gained through my degree into such a large project is fulfilling and feels like the past four years did have a purpose. The best part of the experience was finally getting results and filling pages in

the final thesis paper. Seeing that I was able to produce actual work and results felt like finding the small error in your computer code that was preventing your code from working. Euphoric. The most challenging was keeping to a constant work schedule, especially early in the year where it was mainly literature review and gaining an understanding of the current space of research for my topic. Eventually I got past it, but it was not something I particularly enjoyed, but I do understand the importance of this starting step. My major takeaway from this thesis project, is that I can complete these large, intimidating projects. I learned what I was capable of and gained confidence to pursue whatever it may be in the future. Mostly I want to maintain my work ethic and schedule that I have created for myself through this semester into the future by continuing to use some of the burnout prevention strategies I have learned.

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1. Background

1.1 Mitochondria Mutation and Substitution Rate Paradox

Mitochondria are organelles present in most eukaryotes that perform oxidative phosphorylation within their inner mitochondrial membrane (1). They originated as an alpha-proteobacterial lineage that was probably engulfed by a primitive eukaryotic cell and evolved to become an essential endosymbiont over time. Symbiosis describes the relationship between two separate organisms. The relationship between mitochondria and their respective host cell can be further described as an endosymbiotic relationship, a relationship between two cell types with one cell residing within the other (2). As a result of this evolutionary origin, mitochondria can be thought of as degenerate lineage of intracellular bacteria. These organelles, therefore, uniquely in animals, have their own genome that is independent from the host cell nuclear genome (though the two genomes have co-evolved for over a billion years).

Human nuclear DNA is in a 2N state, meaning there are two copies of the nuclear genome. Mitochondrial DNA (mtDNA), in contrast, have between 2-10 copies of its genome per mitochondrion and an estimated average of 1000 mitochondria within each cell (3). MtDNA is inherited differently than nuclear DNA due to its clonal origin. Nuclear DNA is inherited through the combination of two separate genomes from both parents versus mtDNA being inherited as a subset of the maternal mitochondrial population. The 16 569 base-pair genome has also been shown to have a higher average mutation rate compared to the nuclear genome (3). Our knowledge on the mechanisms of this differential mutation rate is limited even though rate estimates have been applied in a variety of field such as studies involving human disease, species divergence times and forensic medicine (4). One of these limitations is the paradoxical behaviour of the mutation rate estimates found from pedigree studies and the substitution rate estimates

found from phylogenetic studies in the human mitochondrial genome. Pedigree based analyses estimate generational mutation rates between single transmission events (mother to child inheritance). Phylogenetic studies estimate substitution rates of the human mitochondrial genome at a species level scale (human to chimp for example). DNA substitutions, as opposed to mutations, are polymorphisms that become fixated within a population over time. Under the neutral theory of molecular evolution, with no selection pressure, substitution rates per generation become equal to the mutation rates per generation (5). Mutation rate estimates of the human mitochondrial genome from pedigree studies were found to be 2-100 times larger than substitution rate estimates from phylogenetic studies (0.46-2.5 mutations/site/million years and 0.025-0.26 mutations/site/million years respectively) (6). These findings found in previous literature violate the neutral theory of molecular evolution (6).

1.2 Proposed Explanations of the Paradox

There are many potential explanations for this paradox in the mitochondrial genome, most likely being negative selection acting upon the mitochondrial population rather than neutral selection. Negative selection is the removal of deleterious alleles from a population, which would remain undetected in long timescale phylogenetic studies but be detected in generational pedigree analyses (6). The discovery of selection within mitochondrial populations has yet to be uncovered and remains an area of future study. The mitochondrial genome has been shown to have a mutation gradient, which is the phenomenon where different regions within the genome have varying mutation rates (7). These regions are called hyper variable regions (HVR), and in pedigree studies, HVR may present multiple mutations at a single site that would be otherwise missed in phylogenetic studies thus inflating mutation rate estimates. Another hypothesis for the disparity is the difficulty in measuring *de novo* mutations in mother-child transmission events of

heteroplasmic individuals. Heteroplasmy is the genotypic variation within an organism's mitochondria population due to multiple mitochondria being present in cells. Within a single cell, half the mitochondrial population may contain one allele while the other half contains the other. During oogenesis, oocytes take a subset of the mitochondrial variation from the maternal cell population to pass on to their offspring. Problems detecting *de novo* mutations arise when mothers have low heteroplasmy that may remain undetected due to limitations in sequencing, and that small percentage of the mother's mitochondria that is heteroplasmic gets inherited by the child. In this case, we observe a child with a different allele than the mother due to the missed heteroplasmy and is declared a *de novo* mutation. This limitation in heteroplasmy detection could be responsible in inflating mutation rate estimates at the generational level. Many studies have chosen to limit the amount of heteroplasmy in their studies sample set due to the difficulty in weighting the probability of *de novo* mutations or germline inheritance for their rate estimates (6). Studies have suggested using "timescales" to calibrate mutation rate estimates to substitution rate estimates using paleontological data of species divergence (8,9). The "timescales of evolution" model uses Bayesian analysis estimation to connect mutation rates to substitution rates, creating an exponential curve relationship (9). This model assumes a relaxed molecular clock model that allows for the neutral theory of evolution assumption of equal substitution and mutation rates to be lifted (5,9). Through this model, studies have focused on correcting divergent date estimates and have found success in human-human and human-chimp studies (8,9). This theory has yet to be explained or tested in mitochondria population predictive models and has failed to acknowledge aspects of mitochondrial evolution such as heteroplasmy (9).

1.3 Study Rationale

Limitations in mitochondrial evolution research have remained generally unchanged. Many studies have worked around these limitations rather than trying to expand our understanding of the mechanisms revolving around mitochondrial evolution (6). Human mitochondrial disease modeling is one of many applications of mitochondrial evolution that would benefit from more accurate estimates of mutation and substitution rates. Speciation date estimates also benefit from more accurate rate estimates, providing better insight into the origin of species and how evolutionary history has led to organisms today. Mitochondria are one of the only consistencies in within animal cells with low genotypic change over history. Insights into the origin of this relationship will further our understanding of molecular genetics and evolution. Advancements in the applications of mitochondrial evolution have been stunted due to these gaps our understanding. Mechanisms and behaviours of mitochondrial evolutions, such as the pedigree and phylogeny rates of evolution difference, have yet to be fully understood. This study will use bioinformatic techniques and tools to review some of the current theories explaining this paradox. In the end we will present the impact of assumptions used in the “timescales of evolution” model, specifically the assumption that heteroplasmy can be ignored in rate calculations.

2. Objectives

The primary objective of this study is to evaluate some of the potential reasons for the apparent evolutionary rate (the rate of fixation of newly arising alleles or the substitution rate) difference between pedigree-based mutation rate and phylogeny-based substitution rate estimates in human mitochondrial DNA. Our broad objective is to expand the current understanding on the fundamentals of mitochondrial DNA evolution and the implications of phenomenon such as

sequence heteroplasmy, where multiple mitochondrial genotypes are transmitted to each new generation, and systematic issues pertaining to variant calling in mitochondrial DNA studies. To address these objectives, we will largely consider mother-child paired mitochondrial DNA variant data (a simple pedigree with one parent and one child). By addressing current theories for why evolutionary rate estimates vary across timescale of observation (8,9), we hope to lay the groundwork for the development of explicit mechanistic simulations of mitochondrial population genetics, focused on the bottleneck of intergenerational transmission that occurs at fertilization. Such a framework would allow direct, semi-quantitative interrogation of a wide range of hypotheses, and would provide a more solid theoretical foundation for conducting empirical studies in the future.

2.1 Specific Project Aims

This project aims to answer two main concerns brought up from the current literature on mitochondrial DNA evolution. The first concern is how consistent are the previous literature estimates of the mtDNA mutation rate from family pedigrees over several generations, in comparison to estimates from a large number of mother-child mitochondria DNA pairs collected from more recent studies. The second concern is how well pedigree estimates, in general, agree with phylogenetic estimates. We hypothesize that the observed variation (2-100 times difference across timescales (6)) can be explained by a combination of: negative selection reducing the phylogenetic substitution rate over many generations; heteroplasmy in mother-child pedigree, inflating apparent mutation rate estimates by making it difficult to discern whether a sequence difference is due to a *de novo* mutation, or due to inherited variation; and a conflation of polymorphisms (unfixed variations in a population) and substitutions (fixed differences). Phylogenetic methods assume all sequence variations are substitutions, and therefore the

presence of recently arisen polymorphisms are likely to inflate evolutionary rate estimates. Some of these factors are discussed in Sigurdardóttir et al. (6), Ho et al. (9) and Henn et al. (8).

In Specific Aim 1, we will compute the apparent pedigree-based mtDNA mutation rate by collecting mother-child mitochondria DNA pair data from publicly available sources and counting the number of variants between generations. The total number of variants we find in our dataset will be the amount of mutations/site/generation. To compare to substitution rates and previous literature estimates, we will be converting this estimate to mutations/site/million years. We will not remove samples based on minor allele frequency (MAF), allowing for all fractions of heteroplasmy in our dataset, therefore we predict that our rate estimates will be higher than previous literature estimates (6). We will reflect on the ignorance of heteroplasmy in the “timescales of evolution” theory. In Ho et al. (9) and Henn et al. (8), there was little mention of how the model considered heteroplasmic data. We hypothesize that heteroplasmy presents a more prominent limitation to current mitochondrial evolution research and needs to be accounted for and further investigated.

In Specific Aim 2, we will be observing the presence of a mutational gradient in our dataset and comparing to known HVR (7,8). We will also observe the prevalence of a mutation gradient in our curated dataset and whether any mutation gradient could be responsible for differences to literature rate estimates from aim 1. This will be achieved through histogram analysis of the frequency of the position of variants in the mitochondrial genome. The “timescales of evolution” model accommodates for HVR with their relaxed molecular clock model. Our results will confirm the presence of known HVR in our dataset and reveal any regions not consistent with previous literature (7).

2.2 Consideration of Sex and Gender

The inheritance pattern of the mitochondrial genome in vertebrates is predominantly maternal, which is thought to be because of the physiology of male and female gametes (10). As a result, paternal mtDNA is an evolutionary dead-end, as only maternal mtDNA will be passed on to the next generation. This fundamental principal has recently been suggested to be sometimes violated in humans in a prominent study (10). This study, however, has been controversial and we therefore consider this idea as unlikely and unproven. The evolutionary rate of the human mitochondrial genome can only be considered along the maternal lineage. However, mutation in the mtDNA genome occurs in both males and females. In the present study, we therefore track sequence variations arising from mother to child, regardless of the sex of the child. There are no other sex or gender considerations that are apparent. There are no socio-cultural factors that are known to impact rates of mutation and/or substitution.

3. Methods

3.1 Data Curation

To calculate a new pedigree-based mutation rate estimate, we collected mother-child pair mitochondria DNA variant data from four studies with publicly available data (summarized in Table 1). All four studies provided variant called mitochondrial DNA data, labeling associated mother-child variants and MAF.

The first study used massively parallel DNA sequencing with the Ion Torrent Personal Genome Machine (11). Blood samples for this study were taken from 194 mother-child pairs from eastern China, totalling 388 samples and over 14 000 variants which were called using the plug-in variant caller V 5.0 imbedded in the Ion Torrent Software Suite V 5.0.1 (11). In this

study, they detected 33 variants with associated heteroplasmy among 30 of the mother-child pairings (11). The other variants sampled were assumed to have no heteroplasmy due to the study not providing heteroplasmic allele frequency data. Samples that abide by the following conditions were deemed heteroplasmic: MAF of at least 0.1 and observed in at least 2 reads (11). Heteroplasmy detection below this value using the Ion Torrent's high-depth sequencing was deemed too difficult to differentiate between a true variant or a false positive error (11). These 33 variants observed were also the variants that differed between mother and child in this study, and the factor of that difference is reflected in the MAF. They used a general coverage threshold of 40X for all variants found from the samples (11). Heteroplasmic variants were also confirmed through additional Sanger sequencing of HVR 1, 2 and the coding region of the samples (11).

The second data source was a study on transmission bottlenecks and natural selection in human mitochondrial DNA (12). Samples were taken from the cheek and blood of 96 multigenerational families totalling 345 individuals and 249 mother-child transmission events (12). These samples were sequenced using the Illumina MiSeq platform and variants were mapped to the human reference mitochondrial genome in GRCh37, with the Illumina naïve variant caller (12). Heteroplasmic sites were declared as having a MAF greater than 0.01 and at a depth of over 1000X as opposed to the 0.1 frequency threshold used in the first study (12). The study made publicly available called heteroplasmic variants that differed between each mother and child. Similarly, to study 1, the heteroplasmic variants were also verified through Sanger sequencing.

The third study investigated the germ-line bottleneck in inheritance of human mitochondrial DNA (13). Samples were taken from the cheek and blood of 39 mother-child pairs

totaling 156 individual samples (13). The samples were sequenced through Illumina MiSeq and variants were then called using naïve variant caller and variant annotator from Galaxy by mapping to a reference human mitochondria genome (13). Heteroplasmy was described as sites having a MAF of greater than 1% and a read depth of above 1000X (13). Sanger sequencing was also used to validate these heteroplasmsies.

The final study examined mitochondrial DNA heteroplasmy in autism spectrum disorder (14). This study obtained whole exome sequencing data from 1 905 simplex families (one affected individual) from the National Database for Autism Research (14). In total, there were 933 families within this dataset all of which were blood samples (14). The sequenced reads where then processed through alignment to the human reference genome GRCh38 using BWA version 0.7.1 (14). The reads were then filtered with a minimum depth of 40X, heteroplasmsies were defined by having the minor allele observable on both strand reads and that both strand reads have comparable allele fractions and if the median site coverage was above 70X in the total population (14). Sanger sequencing was also used to validate these heteroplasmsies.

All four studies were combined into a single dataset and harmonized by manually extracting parameters from each dataset necessary to calculate a pedigree-based mutation rate. The data values we took into our combined dataset was the study ID (1,2,3 or 4); the family ID; the position of the variant; the major and minor allele; the MAF for the mother and child; mother, child and sibling ID where applicable and the sample type (blood or cheek). The final dataset contained the variants inferred between mother and child samples of all four studies and included estimates of allele frequencies per individual as a proxy for quantifying heteroplasmy.

There are many differences in the methodology of all four studies outlined in Table 1. When combining data from the four studies, there is a risk for batch effects to introduce error

into our mutation rate estimations. Different heteroplasmy detection, variant calling, and sequencing accuracy can all cause mutation rate estimates to differ among studies. In this project we recognize this error, and we formatted our analysis to target direct mother-child comparisons only. Each study provided variant data from both the mothers and their children, as such the bias between studies is limited. Systematic variation exists however, we expect this to be largely orthogonal to the mutation count estimates.

3.2 Mutation Rate Estimates

To calculate the mutation rate from our dataset, we took the total number of variants called, divided by the number of children sampled, divided by the length of the mitochondrial genome, 16 569 nucleotides. A limitation here is that we cannot guarantee that every position of the entire genome was sequenced to sufficient depth for accurate variant calling. However, since most of the studies used high-depth sequencing, the likelihood of this introducing substantial bias is low.

The above noted calculation resulted in estimates of the apparent mutation rate/site/generation in the mitochondrial DNA sequences of our combined dataset. We differentiated blood and cheek samples using the “Sample” column, which resulted in a blood sample mutation rate estimate, cheek sample mutation rate estimate and a total mutation rate estimate. We additionally checked whether there were any significant differences in the overall inferred mutation rate between studies. This is showed in Table 2, where we saw higher mutation rate estimates in studies 2 and 3 due to the lower heteroplasmy MAF threshold in these studies. This was a known limitation of collecting data from various studies, however our analysis of variants in independent mother-child pairs and our heteroplasmy filter allows for the observation of the relationship between rate estimates and a wide range of heteroplasmies.

3.3 Heteroplasmy Filter

To observe the effect of apparent heteroplasmy on the estimated mutation rate in our pedigree data, we implemented a heteroplasmy filter to sort and remove samples across a range of heteroplasmy thresholds. This allowed us to observe the change in mutation rate estimates, as we lowered the allowed heteroplasmy in the data. This filter used the MAF of the child and removed samples above a certain threshold. As we removed samples, we subtracted the number of sites we removed through our filter from the mitochondrial genome length to account for the loss of that site in the dataset. With our filter, we plotted mutation rate estimates as a function of the filter threshold as seen in Figure 1. We performed a similar procedure for the mothers' heteroplasmy and plotted the mutation rate estimate in response to the filter threshold of the mother MAF (Figure 2).

3.4 “Hotspots” Histograms

The prevalence of mutational “hotspot” regions within our pedigree dataset was assessed to examine whether the known mutation rate gradient in the mitochondrial genome could be observed (7,8,15). Regions such as the HVR1, near position 16 000, and the origin of replication of the heavy and light strands, near the displacement-loop HVR, have been found to have an increased mutation rate compared to other regions of the mitochondrial genome (8,15). We created a histogram using R which took the position data of each variant in our combined dataset and plotted it against the frequency of each position. The higher the frequency at a certain region on the histogram, the higher the apparent mutation rate is in that region. We separated the blood and cheek data and created two histograms depicting spatial variation in the frequency of variants in blood and cheek samples.

4. Results

4.1 Mutation Rate Estimates

Three mutation rate estimates were calculated separately for blood samples, cheek samples, and for the combined dataset. For blood and cheek samples, we obtained a rate of 1.3143×10^{-4} and 2.3482×10^{-4} mutations/site/generation respectively. For the entire dataset, we obtained a rate of 2.0935×10^{-4} mutations/site/generation. Literature based estimates look at mutation and substitution rates in terms of mutations/site/million years. In a literature review, many studies found pedigree mutation rates using a generation length of 20 years resulting in rates ranging from 0.43 mutations/site/million years at the low end to 2.6 mutations/site/million years at the highest point (6). Since these studies have been published, the human generation time has been estimated to be closer to 30 years (16) rather than the 20 years used in the past (6). We calculated per million year rates by dividing our original per generation rate by the generation length in years (20 or 30) and then multiplying that resulting number by one million. Using 20 years as our generation length we found the following values: for blood and cheek samples, we found a mutation rate of 6.5716 and 11.7409 mutations/site/million years respectively and a total dataset mutation rate of 10.4675 mutations/site/million years. These values, compared to the literature values, are greater by about 2.5 to 15.3-fold for the blood samples, 4.5 to 27.3-fold for cheek samples and 4.0 to 24.3-fold across the combined dataset. Using 30 years as our generation length brought the mutation rate difference closer to the literature rate estimates (6). For the blood and cheek samples, we found a rate of 4.3811 and 7.82723 mutations/site/million years respectively and a total rate of 6.9783 mutations/site/million years across the entire combined dataset. The blood samples were still greater by 1.7-fold on the low end and 10.2-fold on the high end. The cheek samples were greater by 3.0-fold on the low

end and 18.2-fold on the high end. The total dataset average was greater by 2.7-fold on the low end and 16.2-fold on the high end. All these pedigree-based mutation rate estimates and fold differences using different generation times are outlined in Table 3.

4.2 Effect of Filtering Heteroplasmic Variants

The mutation rate estimates in Table 3 included all samples regardless of the degree of heteroplasmy within those mitochondrial DNA samples. Many studies in the past have excluded samples with a large degree of heteroplasmy due to its limited understanding and unknown mechanisms (6). Heteroplasmic samples cause difficulty in discerning variants as *de novo* mutations, sequence error or missed heteroplasmy in the mother (6). To observe the impact of heteroplasmic data on the calculated mutation rate estimates from our pedigree-based data, we applied a “heteroplasmy filter” which whitelisted data based on the MAF of the sample. For example, a filter of 0.4 would allow samples with a MAF of 0.4 and below, then would estimate the mutation rate from that subset of the data. The filter threshold and the corresponding mutation rate estimate was calculated from a threshold of 0.6 to 0.0 with a threshold change of 0.0001. The results were plotted for blood and cheek samples using the child’s and the mother’s MAF (Figures 1 and 2).

For the child’s data, both the blood and cheek samples mutation rate estimates remained around their respective average values (about 0.0001 and 0.0002 mutations/site/generation respectively) between the heteroplasmy filter threshold of 0.1 to 0.6 (Figure 1). This is to be expected as the histogram of MAF for the child cheek and blood data showed a limited number of samples within this MAF range (Figure 3). Mutation rate estimates in both blood and cheek child samples decreased as the heteroplasmy filter threshold went from 1.0 to 0.0, with the blood samples having an observable hump around a filter threshold of 0.05. Most of the samples were

within the 1.0 to 0.0 MAF range for both the blood and cheek samples (Figure 3). Mutation rate estimates for the blood samples peaked at 1.4182×10^{-4} mutations/site/generation at around a heteroplasmy filter of 0.0488 and was at the lowest at 7.2529×10^{-5} mutations/site/generation at a heteroplasmy filter of 0.0001. The cheek samples peaked at a threshold of 0.4860 with a mutation rate estimate of 2.3530×10^{-4} mutations/site/generation and was at its lowest at a threshold of 0.0001 with a mutation rate estimate of 7.1049×10^{-5} mutations/site/generation.

For the mother's data, similarly to the child's data, both the blood and cheek samples mutation rate remained at their respective average values (about 0.0001 and 0.0002 mutations/site/generation respectively) between a filter threshold of 0.1 to 0.6 with a hump at around 0.1 for the blood data (Figure 2). Similarly, to the child, the largest frequencies of heteroplasms fall below the 0.1 to 0.6 threshold range (Figure 4). Between 0.0 and 1.0 filter thresholds, we see a decline in mutation rate estimates as the filter reaches 0.0. The blood samples at its highest point had a mutation rate estimate of 1.2826×10^{-4} mutations/site/generation at a filter threshold of 0.1012 and its lowest at a mutation rate estimate of 7.0662×10^{-5} mutations/site/generation at a threshold of 0.0000. The cheek samples have its highest mutation rate estimate at 2.3831×10^{-4} mutations/site/generation at a threshold of 0.0009 and its lowest mutation rate estimate of 6.3449×10^{-5} mutations/site/generation at a threshold of 0.0000. In both A and B of Figure 2, we see outlier datapoints close to the 0.0 filter value. After removing these points, the blood minimum and maximum mutation rate estimates remained constant while the cheek sample's maximum mutation rate changed to 2.2176×10^{-4} mutations/site/generation at a filter threshold of 0.4765 with a constant minimum rate estimate.

4.3 Spatial Variation in Apparent Mutation Abundance

In relation to the large disparity between our calculated mutation rate estimates and the literature-based values, we observed the gradient of mutation across the mitochondrial genome within the collected dataset we curated (7). A histogram was created for both blood and cheek samples separately as well as for the total pedigree dataset. We used the position annotation of each different variant called and looked at the frequency of variants in the total population. There were two main observable peaks on all three histograms, one between the 0-1000-nucleotide region and one at the 16 000-16 569-nucleotide regions of the mitochondrial genome (Figure 5). There was one additional peak in cheek samples at the 11 000-12 000-nucleotide region (Figure 5). The blood samples mutation gradient had a higher mutation prevalence at the 16 000-nucleotide region than at the 1000-nucleotide region. The cheek samples mutation gradient had a higher mutation prevalence at the 1000-nucleotide region than at the 16 000-nucleotide region with the 12 000-nucleotide region being lower than both. The total dataset mutation gradient had a higher mutation prevalence at the 16 000-nucleotide region than the 1000-nucleotide region.

5. Discussion

5.1 Why do Rate Estimates Differ?

From our results, we observed an average fold difference of 3.6 on the low end and 22.3 on the high end using a 20-year generation time between our calculated mutation rate estimates to the literature-based estimates (6). Using a 30-year generation time, we saw a fold change of 2.5 on the low end and 14.9 on the high end. These estimates are much higher than we expected. There are a few hypotheses that explain why our rates are larger than previous estimates. The primary reason we suspect is the inclusion of heteroplasmic data in our combined dataset used to

calculate these mutation rates. In many previous studies, highly heteroplasmic data has been removed from these datasets on the grounds that heteroplasmy causes conflicts when trying to determine a generational mutation rate (6). The major problem introduced by heteroplasmy is the inability to differentiate a missed heteroplasmy and a *de novo* mutation (11). Heteroplasmy is difficult to detect at low levels, usually below 0.01 MAF, due to noise produced while sequencing (17). Mothers with low somatic heteroplasmy would remain undetected for their minor allele, which may be inherited in their child. This inherited allele would be viewed as a *de novo* mutation due to the undetected minor allele in the mother. These false positive mutations have been acknowledged as a limitation in calculating mitochondrial mutation rate estimates and the standard practice is to remove samples with heteroplasmy (6). Our unfiltered calculated rate estimates assumed all heteroplasmic alleles present are *de novo* mutations due to this limitation which could explain the dramatic increase in estimated mutation rates. With our heteroplasmy filter, we observed our mutation rate estimates approach literature rate estimates (6) using a 30-year generation time, as allowed heteroplasmy in the calculation was lowered. This result provides evidence for heteroplasmic pedigree mutation rate inflation. This presents a limitation in the theories of mitochondrial evolution models (“timescales of evolution”) as there is no mention of heteroplasmy accommodation (8,9). We propose future work to create a simulation of single generation, single cell inheritance between mother and children to model the probability that a mutation is truly *de novo* rather than inherited from heteroplasmy in the mothers’ gametes. This simulation would need to consider different percentages of heteroplasmic mitochondrial DNA in cells and the distribution of these fractions in different somatic cell types.

5.2 Impact of Children's Heteroplasmy

Heteroplasmy in the sequenced child was a factor that resulted in exclusion of samples in previous literature (6). The difficulty in distinguishing between *de novo* mutations and inheritance from undetected heteroplasmy would lead to samples being removed from studies determining the mutation rate of the mitochondrial genome (6). We used our MAF filter and applied it to our child samples and plotted as shown in Figure 1. Both blood and cheek samples between a filter threshold of 1.0 and 6.0 maintained a constant rate of mutation at their respective average mutation rate (1.3143×10^{-4} and 2.3482×10^{-4}). Below a 1.0 filter threshold, both blood and cheek sample rates of mutation dropped. Blood sample's mutation rate estimates dropped to as low as 7.2529×10^{-5} mutations/site/generation at a heteroplasmy filter of 0.0001. The cheek sample's mutation rate estimates dropped to as low as 7.1049×10^{-5} mutations/site/generation towards a heteroplasmy filter of 0.0001. Both the blood and cheek samples have a mutation rate of about 2.4 mutations/site/million years using a generation time of 30 years, which are within the literature estimate values of 2.6 mutations/site/million years (6). From these results we can conclude that the child mutation rate estimates approach literature estimates as we allow less heteroplasmy in our rate calculations. This suggests that our dramatically increased mutation rate estimates could be a result of the allowed heteroplasmy in our curated dataset, which differed from previous studies (6).

5.3 Impact of Mother's Heteroplasmy

To evaluate further potential limitations in our curated dataset, we observed the mother's MAF in relation to our mutation rate estimates. We plotted the calculated rate of mutation with respect to the allowed heteroplasmy in the data (Figure 2). In Figure 2, we observed consistent rates of mutation when the filter was set between 1.0 to 6.0. This region of the plot stayed near

the average mutation rate for the blood and cheek samples respectively (1.3143×10^{-4} and 2.3482×10^{-4}). Below the 1.0 filter, we saw a dramatic decrease in mutation rates in both the blood and cheek samples. The blood samples dropped to 7.0662×10^{-5} mutations/site/generation as the filter reached 0.0 allowed heteroplasmy. At this lower bound of the plot, we reach a mutation rate of 2.355 mutation/site/million years using a 30-year generation time which fits within the literature estimates for pedigree-based mutation rate (2.6 mutations/site/million years) (6). For the cheek samples, we see the plot drop off to 6.3449×10^{-5} mutations/site/generation as the filter reaches 0.0 allowed heteroplasmy. Similarly, to blood sample estimates, the rate for cheek samples on this lower bound reached 2.115 mutations/site/million years which fits within the bounds of literature estimates (6). The MAF filter has provided evidence that accepting less maternal heteroplasmy changes the calculated mutation rate of our data towards the higher limits of the literature mutation rate (6). Ultimately, however, available data on heteroplasmy may be insufficient to allow accurate identification of true *de novo* mutations because it is evaluated in somatic tissues for both mother and child, and the degree of heteroplasmy in different adult tissues may be substantially different than what is present in the germline. A detailed modelling study is therefore needed to understand the consequences and predictions of different amounts of somatic and germline heteroplasmy on mother-child variant calling in mtDNA.

5.4 Mutation Rate Gradients In our Dataset

Another explanation for the inflated mutation rate estimates in our combined dataset is the mutation rate gradient found in mitochondrial DNA (7). This gradient has been observed to be mediated by regions in the mitochondrial genome spending time single stranded (7). Single stranded DNA has been speculated to be responsible for a large majority of the mutagenesis in large genomes and has been observed to be responsible for about 50% of the mutagenesis in *E.*

Coli genomes (18). To observe the mutational gradient within the mitochondrial genome, we created a histogram of the frequency of the positions of our collected pedigree variant data (Figure 5). We attempted to observe regions within the mitochondrial genome with increased apparent mutation rates and whether these regions have been studied. Novel “hotspot” regions found in our dataset could explain a portion of the increased average mutation rate. Figure 5 shows the distribution of variants in positions across the total dataset, blood and cheek samples. There are two major peaks across all three plots, one at around 1000 nucleotides and another at around 16 000 nucleotides. These two regions of the mitochondrial genome are known HVR. These HVR are studied areas of the mitochondrial genome that are highly polymorphic and have a large degree of variation (19). At 1000 base pairs, this is the HVR for the displacement-loop complex of the mitochondrial genome (15). The peak at 16 000 base pairs contains the region for HVR 1 (8). In the cheek sample histogram, there is a high peak at around 11 000 nucleotides that is unaccounted for in the literature. This region encodes for the NADH dehydrogenase subunit 4, with no known HVR. This peak was not as apparent in the blood and the total dataset, suggesting that tissue dependent mutation gradients in mitochondrial genomes to be further investigated.

6. Conclusions

The objective of this study was to evaluate previously proposed theories on the differential mutation and substitution rate estimates of the mitochondrial genome when doing pedigree-based analyses versus phylogenetic-based analyses. In doing so, we discovered that heteroplasmy has a larger impact on the pedigree mutation rate estimate than previously reported. We observed mutation rate estimates that were 22.4-fold higher when using 20 years as the generation length and 14.9-fold higher when using 30 years as the generation length comparing to the literature estimates of mutation rate in pedigree analyses (6). When limiting the amount of heteroplasmy in

the calculation of the mutation rate, we observed a change towards the higher end of literature mutation rate estimates of 2.6 mutation/site/million years (6). We found this same behaviour when filtering mother and child MAF. Previous studies have excluded data containing heteroplasmy due to the confusion it causes in differentiating heteroplasmic allele inheritance and *de novo* mutation (6). From our results, we suggest that previous estimates may solely be accurate for mitochondrial mother-child pair data with limited heteroplasmy. When assuming all heteroplasmies are *de novo* mutations, estimated rates of mutation become inflated. To work around this limitation in mitochondrial evolution studies, we propose the use of simulation tools to model the behaviour of heteroplasmic inheritance. We could then predict the likelihood of variants being true *de novo* mutations or false positive mutations caused by a child inheriting an allele from an undetected heteroplasmic mother during mother-child transmission events. We also observed the mitochondrial mutational gradient as a potential area of concern in the calculation of mutation rate estimates from pedigree data (7). We observed peaks in the frequency of variants in two HVR, one at 1000 nucleotides and another at 16 000 nucleotides (8,15). We observed another large peak in variant frequency in only cheek samples at around 11 000 nucleotides. This region encodes for a dehydrogenase subunit with no previously recorded HVR and as such, may be a region of future study revealing the potential for tissue sample dependent mutational gradients in human mitochondrial genomes. In terms of previously hypothesized models of mitochondrial evolution (9), heteroplasmy needs to be considered as we have revealed that it has a large effect on mutation rate estimates for pedigree analyses of human mitochondrial genomes. We have laid some of the groundwork for explicit mechanistic simulations of mitochondrial population genetics in the goal of conducting empirical studies in the future.

7. References

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8. Appendix

Table 1: Summary of the four studies used to create the combined dataset in this study. This table outlines the major differences between each dataset, including sample size, sequencing platform, variant caller and the heteroplasmy MAF thresholds.

Study Number	Sample Size	Sequencing platform	Variant caller	Heteroplasmy Filter Threshold
1 (11)	194 mother pairs, 388	Ion Torrent PGM high throughput sequencing platform	Ion Torrent Software Suite (plugin variant caller)	0.1, at least two reads
2 (12)	345 individuals, 96 families	Illumina Miseq	Naïve Variant Caller	0.01 and 1000X read depth
3 (13)	39 mother child pairs, 156 samples	Illumina Miseq	Naïve Variant Caller and Variant Annotator (Galaxy)	0.01 (1%) 1000X read depth
4 (14)	903 families, 2709	Whole exome sequencing	BWA	Minor allele on both strands, comparable fraction, and median site coverage of 70X

Table 2: Mutation rate estimates from each specific study used in the combined dataset. We see larger mutation rate estimates in studies 2 and 3, due to the heteroplasmic minor allele frequency threshold being 0.01 compared to 0.1 in studies 1. Study 4 does not describe an exact minor allele frequency threshold but rather verifies heteroplasmy through read depth, reads on both strands and a comparable minor and major allele fraction.

Study Number	Study Mutation Rate Estimate (mutations/site/generation)
1 (11)	6.6597×10^{-5}
2 (12)	4.7742×10^{-4}
3 (13)	3.8938×10^{-4}
4 (14)	8.0472×10^{-5}

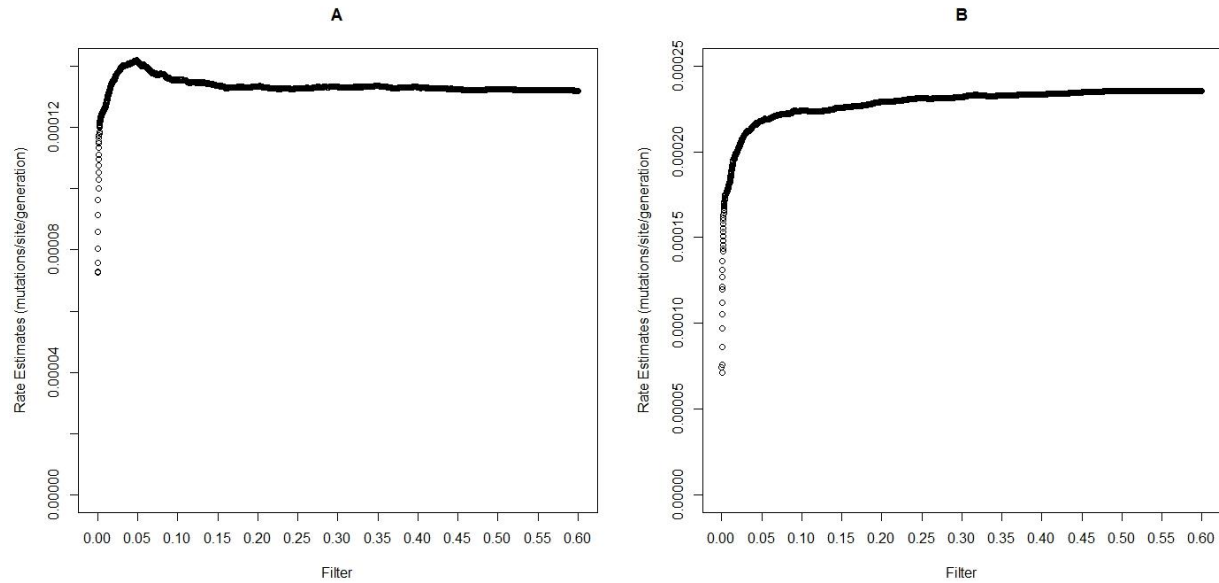


Figure 1: Estimated mutation rate in cheek (A) and blood (B) samples from the combined dataset, using a filtering threshold based on the child's MAF. Variation is expected across tissue types according to differences in the number of cell divisions during development from the single-cell stage until the time of tissue sampling (including relevant stem cell divisions during tissue renewal). Inferred mutation rates in mitochondrial data therefore are not based on observations over a single generation, because both mother and child germline genomes are approximated from somatic tissue sampling. This is likely to be more consequential in mitochondrial studies than in pedigree studies of nuclear DNA.

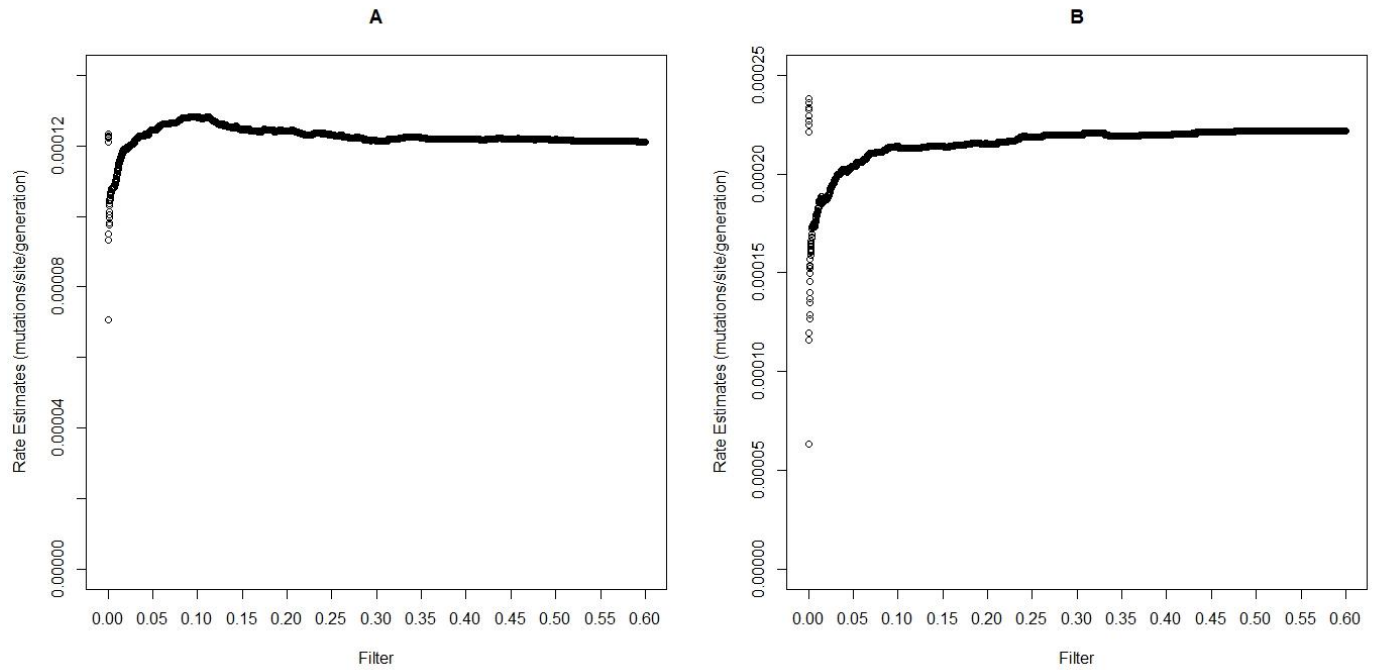


Figure 2: Estimated mutation rate of blood (A) and cheek (B) samples in the combined dataset, using a filtering threshold based on the mother's MAF. The filter represents the amount of allowed heteroplasmic data in the rate estimate calculation.

Table 3: Mutation rate estimates for blood, cheek, and all samples. The third column indicates rate estimates per site per million years with a 20-year generational time, and the fourth column is per million-years with a 30-year generational time. The final two columns display the fold difference in apparent mutation rates compared to the literature highest (2.6 mutations/site/million years) and lowest values (0.43 mutations/site/million years) for both the 20- and 30-year generation time (6).

Sample	Mutation Rate (/site/gen)	Mutation Rate (/site/Myr), 20 year Gen Time	Mutation Rate (/site/Myr), 30 year Gen Time	Fold Difference (20 years)		Fold Difference (30 years)	
				Low	High	Low	High
Blood	1.3143×10^{-4}	6.5716	4.3811	2.5	15.3	1.7	10.2
Cheek	2.3482×10^{-4}	11.7409	7.8273	4.5	27.3	3.0	18.2
Total	2.0935×10^{-4}	10.4675	6.9783	4.0	24.3	2.7	16.2

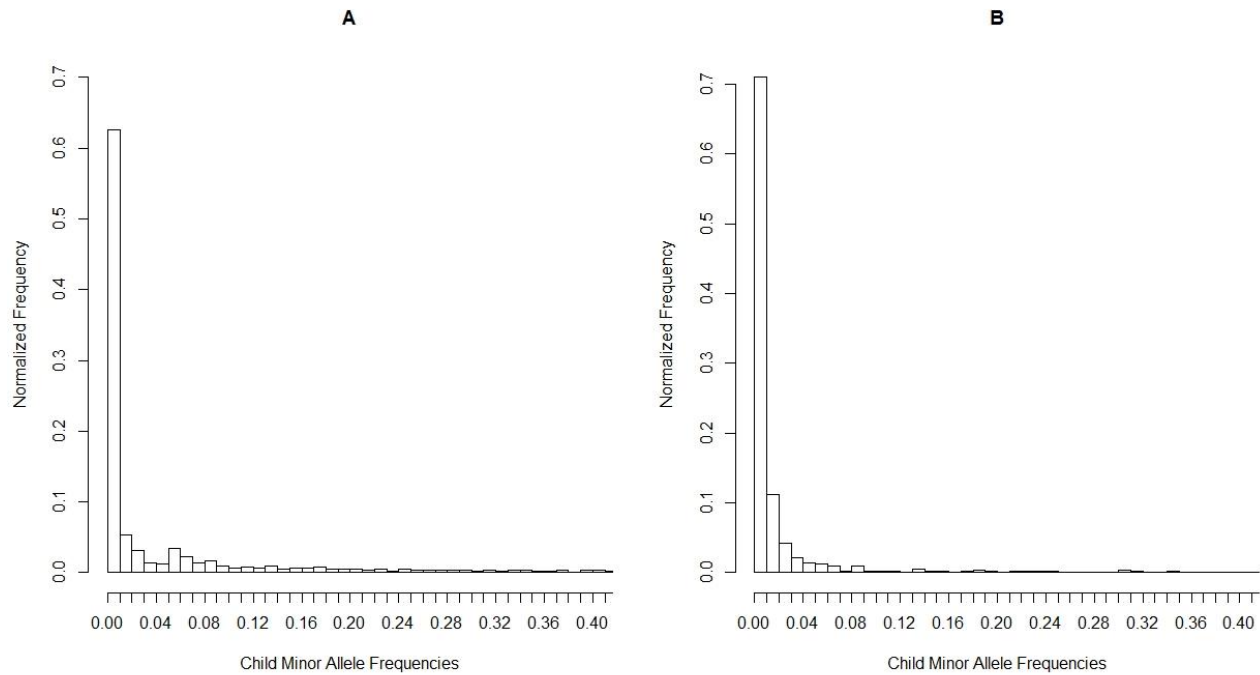


Figure 3: Histogram of the normalized child's minor allele frequencies in the combined dataset in both the blood (A) and cheek (B) samples. Both plots have a large majority of the heteroplasmy within the 0.0 to 0.1 MAF region, however, blood samples (A) have a greater density of variants in the tail of the distribution, indicating greater somatic diversity in blood (A) compared to cheek (B).

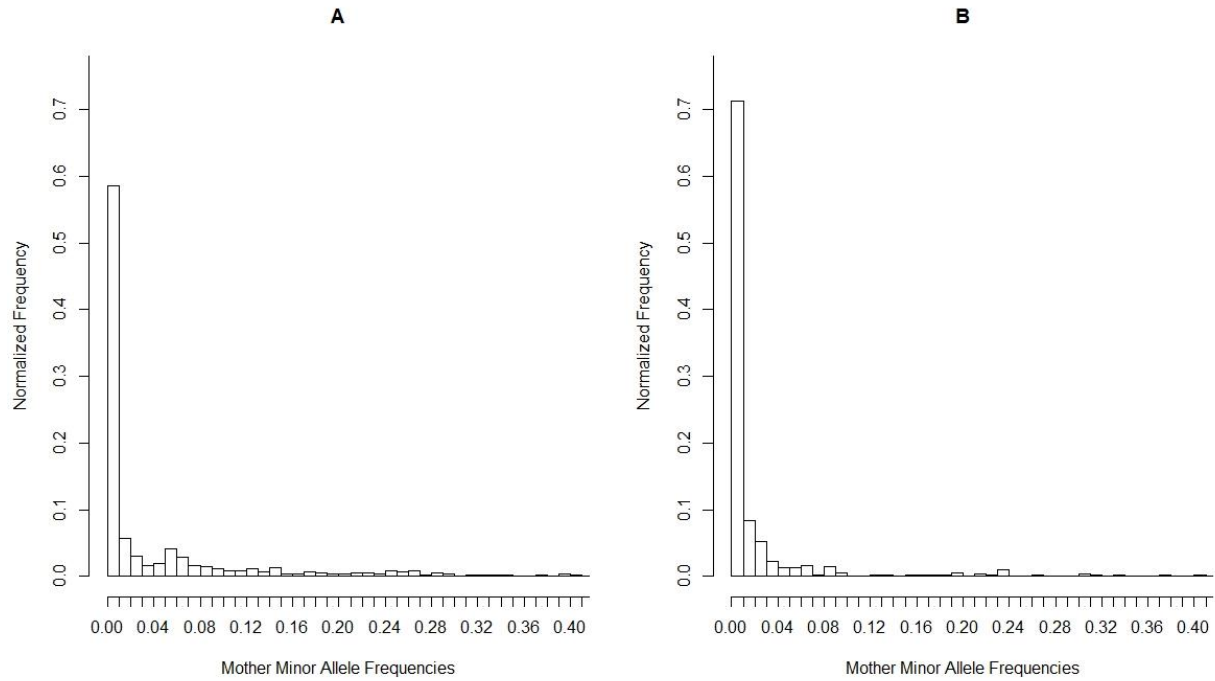


Figure 4: Histogram of the normalized mother's minor allele frequencies in the combined dataset in both blood (A) samples and cheek (B) samples. Both plots have a large majority of the heteroplasmy within the 0.0 to 0.1 MAF region. As in Figure 3, we observe a greater proportion of variants with higher allele frequencies in blood (A) compared to cheek (B).

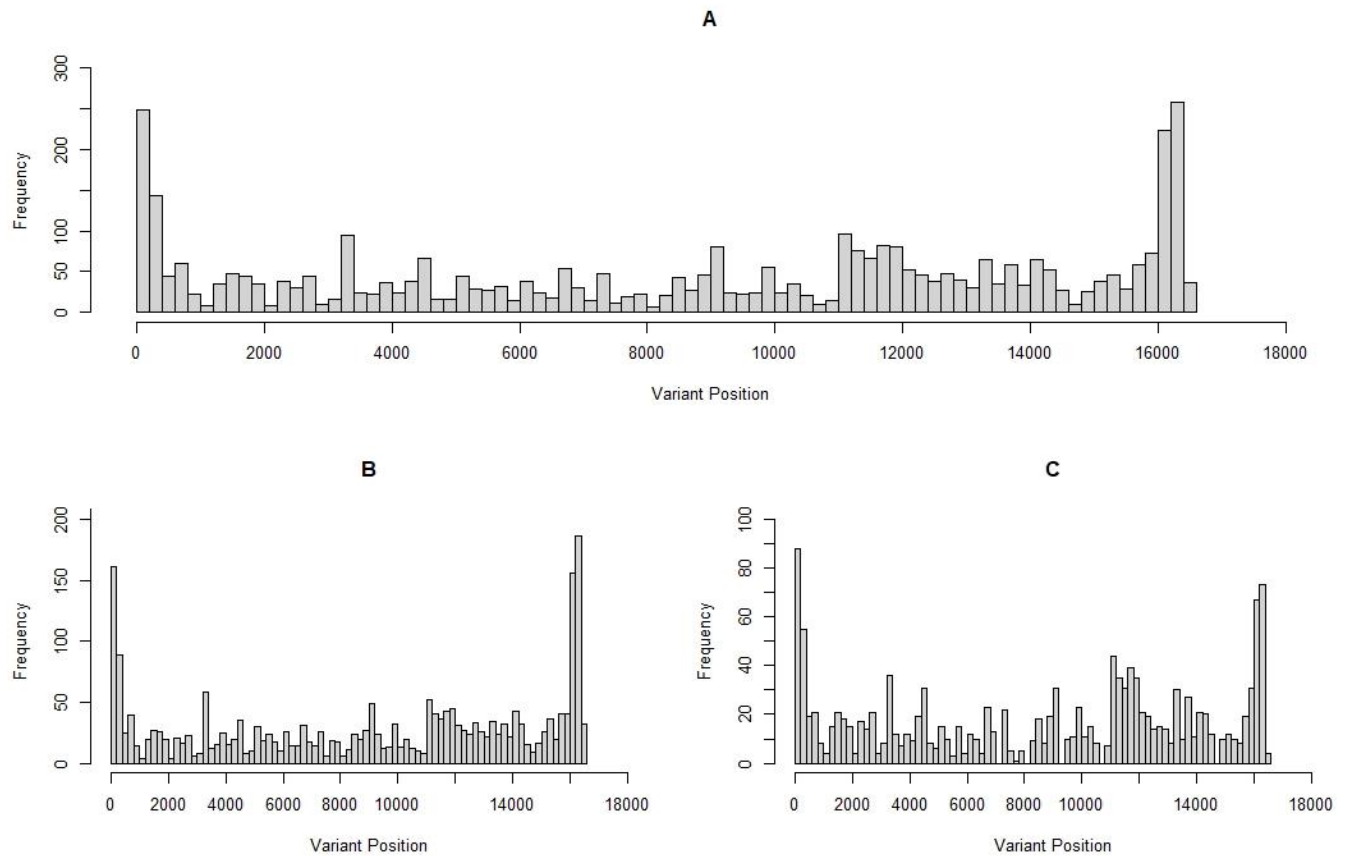


Figure 5: Spatial variation of apparent mutation abundance across the mitochondrial genome.

Panels are: the total combined dataset (A), blood samples only (B), and cheek samples only (C).

There are two prominent spikes within all histograms, at the 16 000 and 1,000 nucleotide regions and a spike in the cheek samples at the 11 000-12 000-nucleotide region (C).